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Securing Canada's AI advantage: A foundational blueprint

16–20 minutes

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Canada – A global leader in AI talent

Canada has long earned its reputation as a global trailblazer in Artificial Intelligence (AI). Since the 1980s, Canada has been a leader in AI, and in some respects is the birthplace of modern AI systems and theories, made possible by the scientific discoveries of Canadian AI researchers. Through advances in algorithms and computing technology, Canada has built a robust and growing AI

ecosystem and industry, known for deep talent and leading-edge innovation.

2018 ACM A.M, Turing Award

(Yoshua Bengio, Geoffrey Hinton and Yann LeCun)

Awarded for conceptual breakthroughs that have made deep neural networks a key component of modern engineering.

This breakthrough has led to major advances in computer vision, speech recognition, natural language processing and robotics and many other applications.

Canada's global leadership comes with a strong reputation for responsible and ethical development of AI. Canada was a founding partner of the Global Partnership on Artificial Intelligence (GPAI) in 2020, a key mechanism to guide the responsible development and use of artificial intelligence in a manner that respects human rights and the shared democratic values of its members. Canada was also the first country to introduce legislation designed to guide AI innovation in a positive direction, and to encourage the responsible adoption of AI technologies by Canadians and Canadian businesses.

Over the past decade, advances in AI have been significant, rapid, and transformative. Canada has been at the forefront of this wave, thanks to investments of over \$2 billion to support AI and digital research and innovation. Canada holds the distinction of being the first country to create a fully-funded national AI strategy – the Pan-Canadian AI Strategy (PCAIS). A first phase of the strategy was launched in 2017, with an investment of \$125 million over five years to strengthen Canada's talent base and global competitiveness in AI research. The Canadian Institute for

Advanced Research (CIFAR), a Canada-based global research organization, was identified as the organization responsible for leading the design and implementation of the strategy's first phase, given the organization's long-standing leadership in advancing AI research.

Since then, CIFAR has been working with Canada's National Artificial Intelligence Institutes—Amii (Edmonton), Mila (Montréal), and the Vector Institute (Toronto)—to attract, retain, and develop top AI talent in Canada. CIFAR has also been working with the institutes to support institutional collaboration, research programs, and training efforts.

To amplify and broaden the gains of the first phase of the PCAIS, and foster responsible adoption and commercialization of AI, the government launched a second phase of the strategy in 2021, with an additional investment of \$443.8 million over ten years, including \$125 million for Canada's Global Innovation Clusters. This second phase of the strategy, combined with the work of the Digital Research Alliance of Canada, gave AI researchers access to advanced research computing resources and promoted the development and adoption of AI standards reflective of Canadian values and interests.

Canada has also taken proactive steps to address the risks associated with AI through its introduction of one of the world's first pieces of AI legislation. The *Artificial Intelligence and Data Act* (AIDA), tabled as part of Bill C-27 (*Digital Charter Implementation Act, 2022*) in June 2022, represents an important first step towards a new regulatory system designed to guide AI innovation in a positive direction, and to encourage the responsible adoption of AI technologies by Canadians and Canadian businesses.

Thanks to these early investments and actions, Canada is well-positioned to compete for its share of the global market which is expected to reach \$1 trillion in size by 2030.

Canada is home to over 1,500 firms specializing in AI, as well as 20 public AI research labs, 75 AI incubators and accelerators, as well as 60 groups of AI investors from across the country. Since 2012, over US\$15 billion has been invested into Canadian AI firms. In this same time period, Canada has generated a number of fast-growing companies deploying or developing AI solutions.

Canada is renowned for its AI ecosystem depth

- **1,500** Canadian companies developing AI solutions
- **3%** of the world's top-tier AI researchers—ranking 6th globally
- **67%** growth in women in AI in 2022-23,
- **highest** year-over-year increase in the G7 for AI talent
- **1st** in the G7 for AI-related scholarly output per capita since 2019
- **57%** increase in AI patent filings by Canadians in 2022- 23
- **US\$15.2 billion** in venture capital attracted to the Canadian AI sector from 2012 to 2023

Budget 2024 and beyond— Growing our AI advantage

AI is already beginning to unlock productivity gains for industries across the economy, and demand is expected to grow as industrial sectors start to integrate AI into their operations. Researchers and companies in Canada are using AI to

create incredible new innovations and job opportunities across all facets of the Canadian economy, from drug discovery to energy efficiency, to new manufacturing techniques and housing innovation.

To capitalize on this global shift and secure Canada's AI advantage, Budget 2024 announced new investments of \$2.4 billion in Canada's AI ecosystem. These investments will help grow Canada's AI Advantage by:

- Providing better access to compute capacity and infrastructure
- Positioning Canada to lead domestically and internationally on responsible and safe AI adoption
- Supporting adoption of AI to drive economic growth and productivity and supporting workers in transitioning to good jobs in the digital economy

Budget 2024 AI Investments at a glance

\$2 billion to launch a new AI Compute Access Fund and Canadian AI Sovereign Compute Strategy.

\$200 million to Canada's Regional Development Agencies to boost AI start-ups to bring new technologies to market, and accelerate AI adoption in critical sectors,

\$100 million for the National Research Council's AI Assist Program to help Canadian small- and medium-sized businesses and innovators build and deploy new AI solutions.

\$50 million to create an AI Safety Institute of Canada to ensure the safe development and deployment of AI.

\$50 million to support workers who may be impacted by AI,

through Employment and Social Development Canada's Sectoral Workforce Solutions Program.

In focus: Scaling up compute access and capacity to meet future demand

AI has incredible potential to transform the economy, improve the way we work and enhance our way of life. Advances in AI, including generative AI, are contributing to transformative changes in scientific research and breakthroughs across a range of industrial sectors, offering substantial economic and productivity advantages to countries that capitalize on the opportunity quickly. Canada's world-class talent pool, research ecosystem, and strong foundation of innovative AI firms are increasingly seeking to leverage these technological advancements to lead scientific discoveries, scale their businesses and grow the industry.

To harness the potential of these new data-intensive technologies and meet anticipated future demand, Canada needs to scale up its compute capacity. Increasing the availability of compute infrastructure in Canada is one of the missing pieces of the puzzle to unlock the potential of Canada's AI ecosystem and capture the full benefits of this exciting, developing technology.

While Canada has long been a leader in the research and development of AI, Canada's partners have been making strategic investments to enhance their computing infrastructure.

Recent international HPC investments

EU	€7.0 billion, European High-Performance Computing Joint Undertaking, a pan-European world class HPC
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	ecosystem.
UK	£900 million in domestic computing capacity (Exascale system at the University of Edinburgh; and AI Research Resource cluster at the University of Bristol)
France	\$283 million USD to fund the \$584 million USD Jules Verne AI Supercomputer, along with EU HPC and Netherlands with installation planned in 2025.

Improving the provision and access to AI compute infrastructure in Canada will help ensure that we grow our domestic firms, and support our researchers and world class AI talent by providing them with the necessary capabilities required to conduct cutting-edge AI research, develop AI models, and ultimately scale companies. Building up Canadian compute capacity may also provide greater opportunities to work with international partners to responsibly unlock the benefits of the technology.

For smaller firms (start-ups and scale-ups), one of the main challenges remains affordability and access to the required compute needed, especially when in competition with larger buyers in the marketplace. Talent and access to compute capacity are the necessary ingredients to building a world class AI ecosystem. To maintain and build on Canada's AI talent advantage, the Government of Canada is taking steps to ensure our organizations, businesses and researchers have access to the compute capacity they need to compete on the world stage and integrate Canadian innovations into the global marketplace and supply chains.

Canada is well-positioned for investments in compute infrastructure, with a world-class clean-energy grid and access to

renewable energy which in turn means there are opportunities to lead in the development of more sustainable compute.

Canada's colder climate and access to abundant sources of water is very attractive to many compute providers, as these are needed to cool modern digital infrastructure. Opportunities also exist for Canada's clean tech sector to work collaboratively with Canadian researchers and companies to help design energy efficient and sustainable data centres and computing facilities.

Major global technology firms and private sector actors are investing heavily in AI and are actively pursuing their own market positions.

Leading estimates indicate that the global market spent US\$46 billion for high-performance servers and generative AI servers in 2023. This is forecast to grow to US\$77.6 billion in 2027, reflecting increasing prices and usage.

National consultations: Toward a Canadian sovereign compute strategy

As Canada strives to strengthen its AI advantage and cement its global leadership position, a strategic and ambitious approach to accelerating access to compute is necessary.

Through Budget 2024, the Government of Canada has committed to consult with key research and industry stakeholders to develop and set criteria for Canadian AI Sovereign Compute Strategy and the AI Compute Access Fund.

Canadian AI Sovereign Compute Strategy and the AI Compute

Access Fund

The Canadian AI Sovereign Compute Strategy will guide Canada's efforts to develop AI compute infrastructure over the long term. The strategy will guide federal investments and actions to build domestic compute capacity that helps grow Canada's AI advantage.

Through the Canadian AI Sovereign Compute Strategy and the AI Compute Access Fund, the Government of Canada will look to take actions that support Canadian AI researchers and developers to access the compute they need in the near-term.

Stakeholders are invited to share their views about how the strategy and the access fund should be designed and implemented.

Consultations will seek input from a number of interested parties including, but not limited to:

- AI Developers
- Universities and Researchers
- Research and Ecosystem Organizations
- AI hubs, Incubators and Accelerators
- Cloud/Data Center Operators
- Semiconductor/Advanced Electronics firms
- Infrastructure Providers and Investors
- Indigenous Groups
- Federal Departments and Agencies

- Provinces/Territories/Municipalities
- International Partners

AI Compute Access Fund

The AI Compute Access Fund will provide direct support Canadian AI researchers and developers to access the compute they need in the near-term.

The guiding principles

As we consult Canadian stakeholders, the Government of Canada has outlined the following general guiding principles for the development of the AI Compute Access Fund and the Sovereign AI Compute Strategy:

1. Identify near-term investment opportunities, acknowledging the immediate needs for increased compute access
2. Establish a base of world-class domestic sovereign AI compute infrastructure that serves as a longer-term foundation for research and innovation, commercialization, prosperity and economic growth
3. Develop a multi-faceted approach that yields results across the Canadian ecosystem, supporting companies, researchers and institutions
4. Employ collaborative approaches that encourage international cooperation
5. Leverage the contributions of the private sector and other levels of government to multiply impact

Different models to accelerate access to compute

To prepare for these consultations, the government has undertaken an initial review of models adopted internationally. These include **Supply Side Approaches** whereby compute infrastructure development is promoted either through direct investment in new infrastructure or indirectly by incentivizing investments through providers of compute infrastructure. **Demand Side Approaches** are also available and focus on leveraging market-based solutions.

In considering its approach, Government will need to determine where it can deliver the most impact. On the research side, increasing the capacity of AI compute currently available to Canada's AI researchers will need to recognize Canada's existing network of digital research infrastructure. This will involve working with the Digital Research Alliance of Canada to identify the needs of the research community and the long term sustainable compute power required to conduct world leading research and enable scientific breakthroughs, including seeking to leverage collaborations with trusted international partners.

Stakeholder views are sought regarding the viability and merits of the following options and potential hybrids of these options to help meet Canada's current and future needs. A multi-faceted approach is likely to employ elements of each of these models to inform the longer-term Sovereign Compute Strategy:

- **Partnering to develop new AI compute infrastructure –** Investments in the development of net-new high performance computing infrastructure through a public-private partnership arrangement. Working hand in hand with industry leaders who

have the expertise in building and operating physical compute infrastructure could translate Canada's investment into increased availability of sovereign compute power for the entire AI ecosystem.

- **Leveraging the market power of AI developers** – Providing funding directly to AI stakeholders or aggregating the demand of AI stakeholders, such as Canada's most promising AI development companies, to help grow the demand for Canadian-based compute capacity. In doing so, public funds would serve to help improve the business case for the private sector to investment in Canadian-based compute capacity.
- **Expand the capacity of compute infrastructure providers** – Providing support to incentivize increases in the supply and availability of compute within Canada. Support could take the form of grants to help offset capital expenditures, or through tax or energy incentives. Investing directly to support providers for compute power in Canada could allow for faster scale-up of capacity and deliver increased compute availability to the Canadian market over the medium-term.

AI stakeholders: Have your say

These **consultations** will be carried-out beginning June 26, 2024 through a variety of means and venues:

- An [online survey](#) for interested stakeholders has been established to submit written comments.
- A series of **in-person and virtual roundtable meetings** with key clusters of stakeholders will be held over the summer.

- **Targeted bilateral consultations**, including with indigenous communities, experts, industry leaders, provinces and territories, and international partners will also be held.

Submissions will be welcomed throughout the summer until September 6, 2024. Please send to aicompute-calculia@ised-isde.gc.ca.

Views will be collected, and the publication of a "What We Heard Report" will follow the consultations to inform the development and implementation of these two initiatives and validate the needs, challenges and opportunities facing the AI ecosystem in Canada.